

Visual Hallucinations in Serotonergic Psychedelics and Lewy Body Diseases

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Background and Hypothesis: Visual hallucinations (VH) are a core symptom of both Lewy body diseases (LBDs; eg, Parkinson's disease and dementia with Lewy bodies) and serotonergic psychedelics (SPs; eg, psilocybin and mescaline). While these conditions differ in etiology, overlapping phenomenology, and neural mechanisms suggest shared pathways. This review explores similarities and differences in VH between LBDs and SPs, focusing on phenomenology, cortical function, and serotonergic modulation.

Study Design: This narrative review synthesizes findings from neurology, cognitive neuroscience, and systems neuroscience to compare VH in LBDs and SPs. The literature includes studies with both human subjects and animal models that examine cortical activity patterns, neuromodulatory mechanisms, and VH phenomenology.

Study Results: Both LBDs and SPs exhibit distinct visual aberrations, ranging from minor metamorphopsias to complex hallucinations. Some features in LBDs resemble those induced by SPs (eg, illusory motion and entity encounters), suggesting shared neural mechanisms. Neuroimaging studies indicate a common pattern of hyperactive associative cortex and hypoactive sensory cortex. At the neuromodulator level, SP-induced VH involves

serotonin 2A and 1A receptor (5-HT_{2A}R and 5-HT_{1A}R) modulation, while in LBDs, 5-HT_{2A} receptor upregulation correlates with increased VH, and its inhibition (eg, with pimavanserin) reduces VH. Two shared cortical signatures are highlighted: reduced visual evoked responses and shifts toward visual excitation.

Conclusions: Examining cortical and neuromodulatory similarities between LBD- and SP-induced VH may elucidate the link between sensory degradation, excitation, and hallucinogenesis. Future research should employ real-time neuroimaging of discrete hallucinatory episodes to identify shared mechanisms and develop targeted interventions for LBD hallucinations.

Key words: Serotonin receptors; excitatory/inhibitory balance; sensory deprivation; visual hierarchy; hallucinogenesis; phenomenology.

Introduction

"His first misperceptions occurred when he was in a night club; the skin of the other dancers, even their faces, seemed to be covered with tattoos. At first, he thought the tattoos

were real, but they started to glow and then to pulse and writhe; at that point, he realized they must be hallucinatory. As an artist and a psychologist, he was intrigued by this experience—but frightened, too, that it might be the beginning of uncontrollable hallucinations of all sorts.”

— from Oliver Sacks’s *Hallucinations*.¹

This quote describes the first hallucination of a patient with Parkinson’s disease (PD), a neurodegenerative disorder characterized by abnormal aggregates of the protein alpha-synuclein, known as Lewy bodies. The location of these aggregates varies and serves as a marker of Lewy body disease (LBD) pathology. In PD, they localize to the brainstem or substantia nigra, while in Parkinson’s disease dementia (PDD) and dementia with Lewy bodies (DLB), they appear in limbic, transitional, or cortical regions.^{2,3} As the quote illustrates, visual hallucinations (VH) also mark LBD pathology.^{4–10} The quote highlights three additional key points motivating this review: (1) simpler hallucinations often precede more complex ones in LBDs,^{11–15} (2) these experiences can distress patients,^{16–19} and (3) they resemble hallucinations induced by serotonergic psychedelics.²⁰

This last point suggests that serotonergic psychedelics (SPs) like psilocybin, mescaline, lysergic acid diethylamide (LSD), N,N-dimethyltryptamine (DMT), and 2,5-dimethoxy-4-iodoamphetamine (DOI) could model hallucinogenesis in LBDs.^{21,22} While SPs act on multiple neuromodulators,^{23,24} their hallucinatory effects are primarily linked to serotonin 2A and 1A receptors (5-HT_{2A}R and 5-HT_{1A}R).^{25–29} The 5-HT_{2A}R is also implicated in LBD hallucinations,^{30–33} which contribute significantly to clinical and economic burden in these patients,¹⁰ and predicts earlier institutionalization, caregiver distress, and reduced quality of life.^{18,19} If validated as a model, SPs could be used to investigate hallucinatory mechanisms, develop biomarkers of hallucination proneness, and test interventions for managing these experiences in LBD patients.

While VH in SPs and LBDs may share some etiological overlap, key differences exist. This review from the International Consortium on Hallucinations Research (ICHR) examines these similarities and differences in phenomenology, cortical activity, and serotonergic modulation. Prior ICHR reviews have compared SP-induced VH with those observed in schizophrenia, which are typically part of a broader, multimodal symptom profile dominated by auditory hallucinations.^{34,35} In contrast, VH in neurodegenerative diseases like LBDs are often unimodal and primarily visual.³⁶ For this reason, our comparison focuses on unimodal VH. We identify relevant cortical signatures (see Onofrij et al.²¹ for a review of thalamic contributions) and highlight evidence that 5-HT_{2A}R and 5-HT_{1A}R exert differential effects across the cortical hierarchy, inducing hyper-frontal and hypo-sensory activation. We also suggest that sensory degradation and

cortical excitation may play key roles in SP- and LBD-induced VH.

Phenomenology

Classifying Hallucinatory Phenomena

To classify hallucination-like experiences, Blom³⁷ proposes the following definitions: (1) *hallucinations*—percepts without external stimuli (eg, a person sitting in an empty chair), (2) *illusions*—misidentifications of external stimuli (eg, a chair misperceived as a crouching person), and (3) *metamorphopsias*—distortions of external stimuli (eg, a stationary chair appears to move). Additionally, LBD research often references *minor hallucinations* that are common during early disease stages.¹⁵ This class can include metamorphopsias, misidentification illusions, and *pareidolia* (ie, seeing meaningful patterns in ambiguous stimuli, such as a face in the clouds), but also *passage hallucinations* (ie, a fleeting, ill-defined entity perceived in peripheral vision) and *presence hallucinations* (ie, the felt sense that an entity is close by). Distinctions are also made between *simple hallucinations* (eg, flashes of light or geometric shapes) and *complex hallucinations* (eg, a whole object or an entire scene). This section compares these classes of phenomenon in LBDs and SPs, highlighting similarities and differences.

Metamorphopsias

Recent research highlights diverse metamorphopsias in LBDs,^{38–46} with distortions reported in approximately 75% of PD patients. These include altered color, shape, or size perception (ie, metachromatopsia, dysmorphopsia, and micro/micropsia), seeing multiples of a single object (ie, di/polyopia), or perceiving stationary objects as moving (ie, kinetopsia). Similar types of distortions are reported with serotonergic psychedelics.^{20,47–49} While largely anecdotal, limited psychophysical studies have explored some of these psychedelic phenomena systematically.^{50–53}

Because visual distortions depend on sensory stimuli, they resemble classic visual illusions,⁵⁴ which can be studied using psychophysical modulation of stimulus and task parameters.⁵⁵ Investigating their origins psychophysically by isolating sensory and perceptual components may be more feasible than studying higher-level hallucinations. Focusing on visual distortions common to both LBDs and SPs could reveal shared neural mechanisms (see [Box 1](#)).

Minor Hallucinations

Minor hallucinations have been reported in 42% of newly diagnosed LBD patients and are the most prevalent hallucinatory symptom of these disorders.^{13,15,67–69} These phenomena include passage hallucinations, misidentification illusions, and pareidolias.^{5,12,15,70} While not

Box 1. Common Hallucinations and Shared Mechanisms

Neurodegenerative and pharmacological processes affect the visual system in significantly different ways, yet shared mechanisms may underlie *specific* hallucinatory phenomena common to both LBDs and SPs. Illusory motion (kinetopsia) exemplifies this potential. It is reported in ~25% of PD patients⁴³ and is a hallmark of psychedelic experiences.^{20,47,49} Convergent psychophysical evidence suggests the involvement of a specific perceptual process in both groups: impairments in higher-order, global-motion integration but not lower-order, local-motion processing.^{56–58} This dissociation implicates the middle temporal (MT) cortex, a region selective for higher-order motion processing. Reduced MT responsiveness is also linked to hallucination proneness in LBD patients.^{59–62} These findings suggest MT hypoactivation may contribute to illusory motion in both groups. Combined psychophysical and neuroimaging evidence could uncover links between other metamorphopsias in LBDs and SPs.

Other hallucinatory phenomena are less viable candidates for this “common phenomenology” approach. Simple hallucinations, such as vivid geometric patterns, are a canonical SP-induced effect.^{20,47,49} In contrast, they are rare in LBDs.⁶³ Moreover, these types of simple hallucinations are commonly the result of eye disease (ie, Charles Bonnet syndrome),⁶⁴ which may be concurrent with LBD diagnoses.⁶⁵ This could confound comparisons of simple hallucinations in LBDs and SPs. An even clearer counter-example to this approach is synesthesia, which is commonly reported following SP administration⁶⁶ and never reported in the LBD literature. Ultimately, identifying shared and distinct hallucinatory mechanisms in LBDs and SPs will depend on the careful selection of which phenomena to study.

emphasized in the psychedelic literature, anecdotal accounts of these phenomenon can be found in online reports of SP-experience: “*I began experiencing pareidolia, seeing faces everywhere, mostly Aztec and Inuit art looking faces, but also some angry looking faces in the bark of trees.*” (pareidolia account from Erowid Experience ID 98005; Published Nov 9, 2022); “*To further complicate matters, plants and rocks constantly morphed into animals and vice versa. I can’t recall exactly how many snakes/tree roots I stepped on or how many squirrels/trees I bumped into.*” (misidentification account from Erowid Experience ID 110580; Published June 3, 2017); “*My eyes were also playing tricks on me. For example, I kept thinking I saw somebody walk by the entrance of my room out of the*

corner of my eyes, and then I would quickly look and nobody was there.” (passage hallucination account from Erowid Experience ID 46857; Published Nov 4, 2005). Moreover, a recent semantic analysis of nearly 40,000 psychedelic experience reports found that peripheral hallucinations (i.e., resembling passage hallucinations) were relatively common across compounds,⁴⁹ despite little prior attention.^{20,47,48} This highlights the need for more quantitative methods to uncover underrecognized yet widespread features of psychedelic experience.

Presence hallucinations, another minor hallucination common to LBDs,¹² occurs in a range of non-clinical contexts,^{71,72} including during bereavement,⁷³ religious practice,^{74,75} and SP administration.⁷⁶ Ethnographic research on shamanic uses of ayahuasca found that cultural and contextual expectations can shape the identity of the perceived presence.^{77,78} This effect of expectation parallels results from LBD research, were approximately 50% of patients recognize the identity of a felt presence.¹²

Complex Hallucinations

In LBDs, complex hallucinations occur at rates between 22% and 38%¹³ and often involve individual entities (objects, animals, or people) but can occasionally transform the entire visual scene.^{79–81} To a first approximation, they resemble SP-induced complex hallucinations^{47,82–84} though SP content can be more fantastical,⁸³ whereas LBD content tends to be mundane though sometimes distorted.⁸¹ It is unclear if these differences reflect distinct neural mechanisms or contextual expectations.^{77,78}

Even different SP compounds induce different visual effects,⁴⁹ with entities more commonly encountered during DMT experiences compared with others.²⁴ Strassman (1995) reported that 50% of participants encountered entities after high-dose DMT, a figure supported by larger studies describing encounters with guides, spirits, or aliens.^{85–89} This propensity of entity-inducing compounds may also result from either distinct neural mechanisms or contextual expectations. Systematic interviews with LBD patients who have experienced SP-induced hallucinations prior to their diagnosis, ideally from a range of SP compounds, could provide valuable insights into these phenomenological similarities and differences (see Suzuki et al.⁹⁰ for a model-based approach).

A Hypothetical Common Pathway?

In LBDs, the emergence of visual distortions and/or minor hallucinations often, though not always, signals the start of a disease progression toward more complex hallucinations. A similar progression from simple to complex VH can occur with SPs, though over hours rather than years.⁴⁸ Curiously, sleep deprivation follows a comparable trajectory, transitioning from visual distortions to complex phenomena over days.⁹¹ Future research might explore whether these trajectories reflect superficial

Table 1. Possibility of Comparing Hallucinatory Phenomenology in LBDs and SPs

Perceptual effect	Lewy body diseases	Serotonergic psychedelics	Comparison
Metamorphopsias	Commonly reported ^{38–46} Prevalence (PD): ~75% ⁴⁴	Commonly reported ^{20,47–49,82} Prevalence: NA	Possible
Synesthesia	Not reported Prevalence: NA	Commonly reported ^{20,49,66} Prevalence: ~50% ⁶⁶	NA
Simple Hallucinations	Rarely reported ⁹² Prevalence: NA	Commonly reported ^{20,47–49,82} Prevalence: NA	Difficult
Pareidolia	Commonly reported ⁴⁶ Prevalence (PD): ~14% ⁴⁶	Rarely reported Prevalence: NA	Difficult
Misidentification Illusions	Commonly reported ^{15,43,46} Prevalence (PD): ~20% ⁴⁶	Rarely reported ⁸² Prevalence: NA	Difficult
Passage Hallucinations	Commonly reported ^{15,45,46} Prevalence (PD): ~46% ¹⁵	Rarely reported ⁴⁹ Prevalence: NA	Difficult
Presence Hallucinations	Commonly reported ^{15,45,46} Prevalence (PD): ~25% ¹⁵	Rarely reported ^{77,78} Prevalence: NA	Difficult
Complex Hallucinations	Commonly reported ^{5,6,13,79,80} Prevalence (PD): ~22% ⁵	Commonly reported ^{20,49,82,83,85–87} Prevalence (DMT): ~45% ⁸⁶	Possible

Table 1 outlines which hallucinatory phenomena are reasonable candidates for comparison between LBDs and SPs. While quantitative prevalence estimates are often available for LBDs, they remain sparse in the largely qualitative SP literature. To avoid artificial equivalence, the table presents both literature-based designations (“commonly,” “rarely,” or “not” reported) and quantitative estimates where available. These labels reflect either empirical strength or consistent anecdotal reporting, as supported by citations. A comparison is deemed “possible” when a phenomenon is commonly reported in both groups, “difficult” when common in one but rare in the other, and “not applicable” when absent from one group entirely.

similarities or reveal shared mechanisms that naturally progress from lower-order to higher-order regions of the visual hierarchy, despite unfolding over different time-scales. In general, careful comparisons between LBD and SP hallucinations, such as those outlined in **Table 1**, may help identify which perceptual phenomena reflect shared underlying mechanisms and which are likely to diverge across clinical and pharmacological contexts.

Cortical Function

Differential Effects Across the Cortical Hierarchy

SPs consistently cause differential effects across the cortical hierarchy.⁹³ The first modern neuroimaging study⁹⁴ used single photon emission computed tomography (SPECT) to measure regional cerebral blood flow (CBF), showing mescaline-induced hyper-frontal and hypo-sensory metabolic changes (see **Figure 1A** for specific regions affected). Similar patterns have been observed using PET,^{29,95,96} MRI-derived CBF measures,⁹⁷ and fMRI global signal topography **Box 2** (see **Figure 1B**).⁹⁸ These patterns of hyper- and hypo-activity align with a cortical divide between “intrinsic” (internal cognitive/affective) and “extrinsic” (sensory/perceptual) systems that are distinguished by gene expression⁹⁹ and functional connectivity.¹⁰⁰

In LBDs, studies report hypermetabolism in frontal regions¹⁰¹ and hypometabolism in primary and associative visual areas^{70,102–106}. These effects distinguish LBD from non-hallucinatory neurodegenerative disorders (e.g., Alzheimer’s disease)¹⁰⁶ and patients without

hallucinations (**Figure 1C**)¹⁰². This differential activity may result from patterns of cortical neurodegeneration that are more pronounced in hallucinating patients. A recent mega-analysis¹⁰⁷ found hallucinating Parkinson’s patients showed reduced cortical thickness in visual and frontal regions (i.e., occipitotemporal, parietal, and frontal cortex) and additionally reduced cortical surface area in occipital/occipitotemporal regions. Finally, resting-state fMRI dynamic causal modeling recently revealed that hallucinations in both LBDs¹⁰⁸ and SPs¹⁰⁹ were linked to reduced bottom-up and increased top-down effective connectivity. These findings align with theories suggesting that sensory deficits and associative hyper-activity contribute to hallucinations across contexts, including neurodegeneration^{35, 110}, sensory deprivation¹¹¹ and meditation¹¹².

Diverse Effects in Associative Cortex

Recently, Collerton et al.¹¹³ developed a framework (**Figure 1D**) to synthesize 8 leading models of VH, including several designed to model LBD hallucinations explicitly. Across the models, the contributions of top-down, associative processes vary considerably. According to different models, hallucinogenesis is attributed to various combinations of *attention* (Attention and Perception Deficit, APD; Activation-input Modulation,¹¹⁴ AIM; Attentional Networks,^{115,116} AN; Active Inference,¹¹⁷ AI; Thalamocortical Dysrhythmia Default Mode Network Decoupling,¹¹⁸ TDDMND), *expectation* (AN, AI, TDDMND), *memory* (AN, TDDMND, Reality

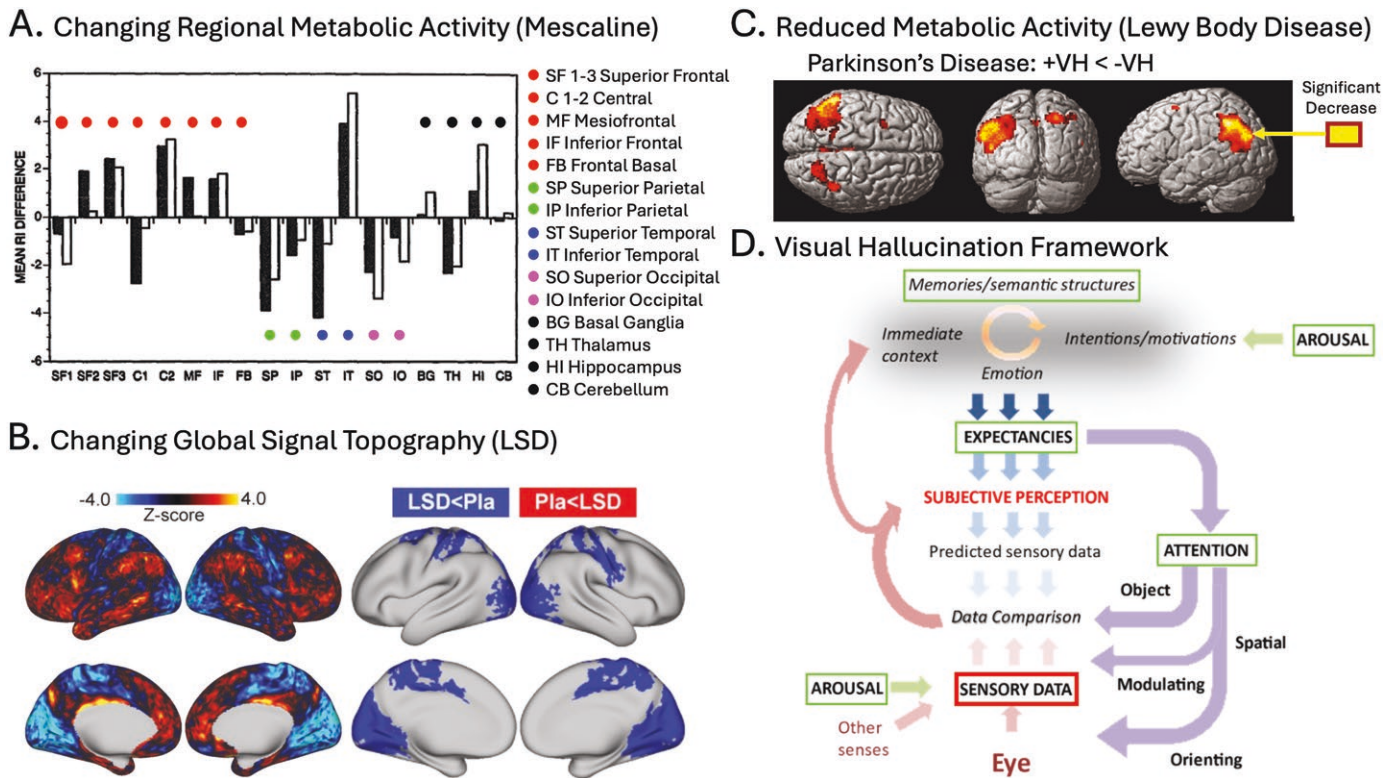


Figure 1. Differential effects across the cortical hierarchy. (A) From Hermle et al.,⁹⁴ showing mescaline-induced changes in regional metabolic activity (SPECT-CBF) compared to baseline. Metabolic activity predominantly increases in frontal regions and decreases in parietal, temporal, and occipital areas, with the notable exception of inferior temporal cortex (IT). (B) From Preller et al.,⁹⁸ showing LSD-induced changes in the fMRI global signal (GS) topography compared to placebo. The left images (unthresholded) depict GS modulation decreases in sensory-motor cortex (cold colors) and increases in the association cortex (warm colors). The right images (type-1 error corrected) depict significant GS reductions in sensory-motor cortex. (C) From Boecker et al.,¹⁰² showing reductions in regional metabolic rate of glucose consumption (FDG-PET) in the occipitotemporoparietal region of hallucinating PD patients (+ VH) compared with non-hallucinators (-VH). (D) From Collerton et al.,¹¹³ showing a framework for comparing across VH models. The image shows the range of cognitive, affective, perceptual, and sensory processes included in eight different models of VH. Different models posit that different combinations of top-down processes (ie, memories, expectancies, arousal, or attention) contribute to the production of VH, but all agree that degraded bottom-up sensory information (ie, sensory data) plays a key role in hallucinogenesis. Source: Data from Hermle et al., 1992, is reprinted from *Biological Psychiatry*, Vol. 32, Issue 11, with permission from Elsevier. Data from Boecker et al., 2007, is reprinted with permission from *JAMA Neurology*. Data from Preller et al., 2018, is reproduced under a CC BY 4.0 license from *eLife*. Data from Collerton et al., 2024, is reproduced under a CC BY 4.0 license from *Neuroscience and Biobehavioral Reviews*.

Monitoring,¹¹⁹ RM), and arousal (AN), or suggest wide-spread network disfunction (ie, the Hodological model),¹²⁰ or specify no role for top-down processing (ie, the Deafferentiation model).¹²¹

In SP research, similar model diversity exists. Divergent accounts of SP function may stem partly from an over-emphasis on resting-state fMRI designs.^{122,123} During rest, SPs induce unpredictable mental states that can evolve rapidly, likely engaging different cortical networks. Combined with small sample sizes and other idiosyncratic factors, different studies observing different network activation patterns may have inspired different models. A biproduct of this diversity means evidence can be selectively compared, within and between LBD and SP studies, that aligns with or contradicts various models. For instance, a study supporting the Attentional

Networks model¹²⁴ found increased coupling between the default mode network (DMN) and the ventral attentional network (VAN) in hallucinating LBD patients. SPs can similarly induce DMN-VAN connectivity.^{109,125} However, the same study also found stronger connectivity within DMN nodes in hallucinators, a consistent finding in LBD research.¹²⁶⁻¹²⁹ This contrasts with SPs, which typically reduce within-network connectivity, including within the DMN, while increasing between-network connectivity.¹³⁰ Reduced within-DMN connectivity is central to one leading theory of psychedelic function,¹³¹ though not all.¹²²

Compared with resting state designs, task-based paradigms can target specific top-down functional networks, constraining comparisons across hallucinatory contexts. Moreover, established neurocognitive tasks

that show reliable behavioral effects in one hallucinatory context could be validated in another. For example, perception of ambiguous stimuli¹²⁴ and pareidolia^{63,70} correspond with LBD hallucinations and could be used to investigate attentional selection and expectation in SPs. In SPs, visual working memory dysfunction is well established^{132,133} and may overlap with findings on LBD hallucinations.^{127,134} Furthermore, task-based approaches are well placed for bridging clinical and preclinical research.¹³⁵

The diversity of hallucinatory models raises the additional question of whether VH across contexts and types (eg, clinical/pharmacological or metamorphopsias/entities) share a common mechanism or emerge from multiple independent pathways. Collerton et al.¹¹³ propose a middle ground between these two extremes that highlights a key point in the hallucinogenic process: the comparison of degraded sensory data with internal expectations. Evidence that a *specific* top-down, associative process is contributing to hallucinogenesis (eg, one generating expectations via memory or comparisons via attention) may vary depending on the specific hallucination context or type. However, across hallucinatory contexts and phenomena, there is evidence that degraded bottom-up, sensory information plays a key role in hallucinogenesis.

Reduced Visually Evoked Activity

Degraded visual information is a unifying feature of every model reviewed by Collerton et al. (2024). The failure of the visual system to encode and/or represent high-fidelity sensory information may drive various top-down processes to “fill the gaps,” resulting in hallucinatory experience. In LBDs, visual dysfunction is well established.¹³⁶ It can result from a range of ocular and visual disturbances¹³⁷ and corresponds with minor hallucinations more so than complex hallucinations.¹³⁸ Multiple genetic markers have been linked to these visual deficits, as well as to reductions of both metabolic activity and blood flow in posterior brain regions in these patients (see Weil et al.⁶⁵ for a comprehensive review).

One functional signature of degraded visual processing in LBDs is reduced stimulus-driven activity in visual regions. fMRI studies have linked LBD hallucinations to reduced activity in occipital¹³⁹ and temporal visual cortices,¹⁴⁰ as well as higher-order visual regions like the superior parietal lobule and the frontal eye fields.¹²⁴ Multiple studies have found reduced activation in the higher-order motion area MT in LBD patients.^{59–62,124} Recently, Vignando et al.¹⁴¹ show that the magnitude EEG evoked response potentials (ERPs) in hallucinating LBD patients are notably degraded, compared with non-hallucinators (see [Figure 2A](#)).

SPs also reliably reduce stimulus-evoked activity in visual regions. In multiple EEG studies,^{142,143,145,146} SP administration suppressed the N170 ERP. The N170

is a negative potential that occurs ~170 ms after stimulus onset and is typically associated with the perception of complex visual objects, faces in particular.¹⁴⁷ Crucially, this signature of visual suppression correlated significantly with subjective measures of VH ([Figure 2B](#)) and is reversed by blocking the 5-HT_{2A} receptor (see the Serotonin Receptor section for more details).¹⁴³ Additionally, fMRI evidence of SP-reduced stimulus-driven activity has been observed in higher-order visual regions (i.e., fusiform gyrus and inferior temporal gyrus),¹⁴⁸ and observations with both human fMRI¹²⁵ and mouse two-photon calcium imaging (2P-Ca²⁺)¹⁴⁴ studies find SPs reduce stimulus-driven activity in primary visual cortex (V1; [Figure 2B, C](#)). While there is a clear correlation between SP-induced visual suppression and hallucinations, it remains to be shown whether this suppression plays a direct, causal role in SP hallucinogenesis (see [Box 2](#)).

Hyperexcitability in Visual Cortex

Perhaps surprisingly, hyperexcitability in the visual cortex, in addition to reduced sensory activation, is associated with VH in both LBDs and SPs. In LBDs, multiple lines of evidence link hallucinations to visual cortex hyperexcitability. Taylor et al.¹⁶³ found hallucination severity correlated with phosphene thresholds in LBD patients. Hallucinators experienced more complex percepts, suggesting a link between excitability and hallucinations. In a follow-up fMRI study,¹⁶⁴ phosphene thresholds positively correlated with BOLD activity in visual areas in healthy controls but negatively in LBD patients, indicating a loss of inhibitory control. Reduced GABAergic inhibition in hallucinating patients, observed in postmortem studies¹⁶⁵ and magnetic resonance spectroscopy,⁵⁹ further supports increased excitability. Similar changes occur following visual deprivation, where lower GABA levels shift the excitatory/inhibitory (E/I) balance toward excitation.¹⁶⁶

In SPs, empirical evidence of hyperexcitability in the visual cortex is primarily derived from suppressed oscillatory power in the alpha bandwidth (8–13Hz) in EEG/MEG signals. Alpha power modulates cortical excitation¹⁶⁷ and perceptual sensitivity.¹⁶⁸ Alpha suppression was observed in the very first psychedelic neuropharmacology study¹⁵⁴ and remains a robust correlate of SP administration.^{143,169–176} Recently, Muthukumaraswami and Liley¹⁷⁷ showed SP-induced alpha suppression correlates with a proxy for E/I balance (i.e., flattening of the 1/f slope in the 30–50Hz range),¹⁷⁸ indicating a shift toward greater excitation. Whether such excitatory shifts directly trigger hallucinations, cause their persistence, simply increase their likelihood, or are an entirely unrelated phenomenon remains unknown. Addressing this requires experiments measuring SP neural signatures during the

Box 2. Capturing the Onset, Persistence, and Resolution of Hallucinations

Hallucinations occur episodically, not continuously. It is critical to identify features of the hallucination-prone brain when hallucinations are present and absent.¹¹³ Identifying features when hallucinations are absent, called “trait-based” research, could aid in developing diagnostic markers for early identification or management of hallucination susceptibility. “State-based” research emphasizes measuring neural signatures during hallucinatory episodes through hallucination capture experiments. Capture experiments have the potential to isolate dynamical neural processes responsible for transitions between veridical and hallucinatory perception, including the onset, persistence, and resolution of hallucinations.¹¹³

To date, hallucination capture experiments have identified a clear signature of *persistence*, potentially related to increased excitability in visual cortex: hyperactivity in category-selective visual cortices that match hallucinatory content. For example, hyperactivation in speech-related regions corresponds to auditory verbal hallucinations in schizophrenia,¹⁴⁹ while hyperactivation of fusiform gyrus corresponds to face hallucinations in Charles Bonnet syndrome.¹⁵⁰ Dujardin et al.¹²⁶ reported similar effects during LBD hallucinations. While category-selective cortical activation serves as a marker of hallucinatory persistence, capturing signatures of the transition between veridical and hallucinatory perception is more challenging.^{151,152} Clinical hallucinations occur infrequently and unpredictably, making it difficult to systematically measure transient events like *onset* and *resolution*.¹¹³

SPs, on the other hand, reliably induce hallucinations that evolve dynamically, repeatedly emerging, persisting, and dissipating during the drug time course.⁴⁸ While this dynamic context creates its own challenges, it also provides researchers with more opportunities to capture signatures of transitions between veridical and hallucinatory perception. Well-established effects of SP administration, such as visual activity suppression and cortical excitation, should be explored as candidate correlates of hallucinatory transitions.

Surprisingly, no SP hallucination capture experiments have been performed using modern neuroimaging methods. While many resting-state fMRI studies exist,¹²³ including one exploring putative hallucinatory mechanisms,¹⁵³ none track onsets or resolutions of specific episodes. Early SP researchers attempted to identify real-time EEG correlates of hallucinations,^{154–156} reporting that alpha-power suppression corresponds to hallucinatory periods. However, other early studies report significant individual variations in this effect.^{157–162} Modern SP-capture experiments could resolve these inconsistent observations.

onset, persistence, and resolution of hallucinatory events (see Box 2).

Psilocybin has also been shown to amplify the P100 ERP, a positive potential ~100 ms after stimulus onset that is thought to reflect early processing of visual input.¹⁷⁹ Across the four studies that found N170 suppression (reviewed above), there was disagreement regarding effects on the P100. Two studies employing higher-order stimuli (i.e., faces) reported no effect on the P100,^{145,146} while two studies using lower-order stimuli (i.e., Kanizsa stimuli), reported marginal P100 facilitation constrained to medial-occipital electrodes.^{142,143} This stimulus selective facilitatory effect suggests that 5-HT_{2A} R activation may be excitatory earlier in the visual hierarchy and suppressive downstream. Crucially, this is at odds with observations from animal electrophysiology in primary visual cortex,^{144,180,181} indicating that more work is needed to resolve this discrepancy.

It is important to note that hyperexcitability in frontal cortex also likely contributes to LBD and SP hallucinogenesis via feedback circuitry.^{30,108,109} Such feedback may drive the diverse set of top-down mechanisms proposed by the models discussed above. While it is crucial to understand such contributions, in this review we suggest that bottom-up signatures may offer clearer targets for cross-group comparison.

Balancing Suppression and Excitation?

Of the eight models reviewed in Collerton et al. (2024), the Deafferentation Model most directly addresses the paradoxical link between degraded sensory input and visual hyperexcitability. This model was proposed to explain hallucinations in eye disease (ie, Charles Bonnet Syndrome) and sensory deprivation,^{64,121} and suggests that visual deprivation disconnects ascending inputs from upstream cortices, causing compensatory hyperexcitation. Recent work links this disconnection process to destabilized homeostatic plasticity and a shift in cortical E/I balance toward excitation.^{182,183} As SPs cause both reduced visual activation and cortical excitability, they may be useful for investigating links between disturbed homeostatic processes and hallucinogenesis.

Intriguingly, dysregulation of related homeostatic mechanisms contribute to visual plasticity during developmental critical periods.^{184,185} Critical period plasticity has been implicated in the effects of SP-administration,^{186–188} and recent theoretical work suggests CP plasticity plays a role in Hallucination Persisting Perception Disorder.¹⁸⁹ Thus, investigating SP-induced visual suppression and E/I modulation might reveal correspondences between homeostatic dysregulation,¹⁸² critical period plasticity,¹⁸⁴ and hallucinogenesis.¹⁸³

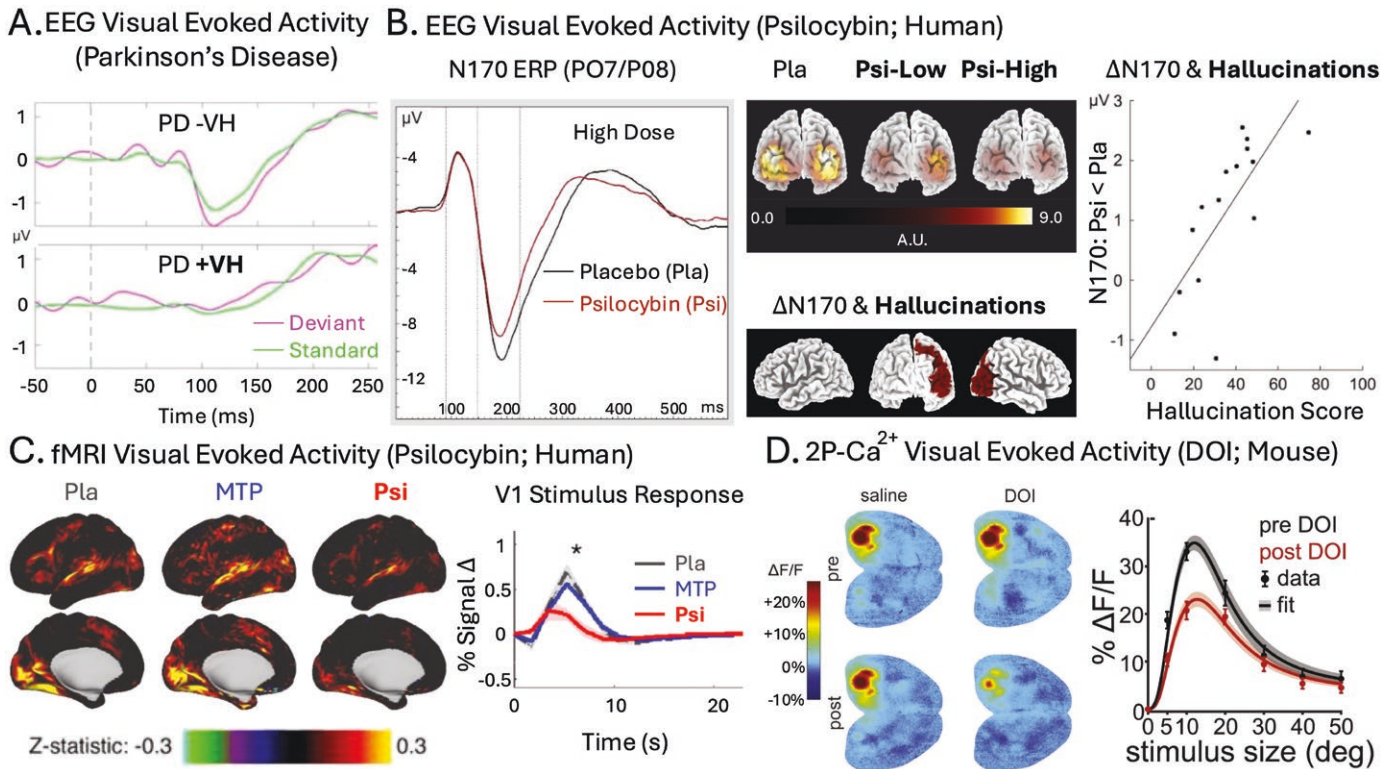


Figure 2. Reduced Stimulus Evoked Response. (A) From Vignando et al.,¹⁴¹ showing averaged EEG visually evoked activity in hallucinating (+VH) and non-hallucinating (−VH) Parkinson's disease (PD) patients evoked during a mismatched negativity task (MMN). In +VH, no negative evoked response potential (ERP) is apparent, while one is clearly evident in −VH for both trial types. (B) From Komater et al.,^{142,143} showing averaged EEG visually evoked activity during psilocybin (Psi) and placebo (Pla) administration. Left panel depicts the suppression of N170 evoked response potentials (ERP) during a high-dose Psi condition. Middle panels show that this suppression is dose-dependent (Psi-Low and Psi-High) and localized to posterior cortex (top images), and that reduced activity in right-posterior cortex correlates significantly with measures of SP-induced hallucinations (bottom images). Right panel shows a similar correlation between hallucination scores and N170 suppression from a later study, where 5-HT_{2A} R blockade was shown to block both Psi-induced N170 suppression and hallucinations. (C) From Siegel et al.,¹²⁵ showing fMRI visual evoked activity during Psi, Pla, and methylphenidate (MTP) administration in humans. Left images depict cortex-wide effects of visual stimulation and right image depicts time course of average voxel activity in V1. Both demonstrate suppressed visual activity during Psi compared to control conditions. (D) From Michael et al.,¹⁴⁴ showing widefield 2P-Ca²⁺ visual evoked activity during DOI and saline (Sal) administration in mice. Left images depict cortex-wide effects of visual stimulation pre and post, with DOI suppressing visual activity in V1 post administration. Right image depicts stimulus-size tuning curves for individual V1 neurons (layer II/III), with DOI again showing a suppressive effect. *Source:* Data from Vignando et al., 2022, and Siegel et al., 2024, are reproduced under a CC BY 4.0 license. Data from Komater et al., 2011, is reprinted from *Biological Psychiatry*, Vol. 69, Issue 5, with permission from Elsevier. Data from Michael et al., 2019, is reprinted from *Cell Reports*, Vol. 26, Issue 13, with permission from Elsevier.

Serotonin Receptors

5-HT_{2A}R & 5-HT_{1A}R Comodulate Hallucinations

Activation of the 5-HT_{2A}R receptor is essential for SP-induced hallucinogenesis. Vollenweider et al.²⁹ showed that blocking 5-HT_{2A}R with ketanserin dose-dependently reduced and eventually abolished psilocybin's hallucinogenic effects. Other SPs show similar modulation.^{26,28} The 5-HT_{1A}R receptor also plays a role. Its blockade intensifies DMT hallucinations²⁷ and its activation attenuates the hallucinatory effects of psilocybin.²⁵ These findings suggest 5-HT_{2A}R and 5-HT_{1A}R co-modulate SP-induced hallucinations.

Parkinsonian disorders are generally associated with loss of dopaminergic function and LBD hallucinations are

closely associated with cholinergic modulation (see Box 3). However, there is mounting pharmacological evidence that interfering with 5-HT_{2A}R activity reduces hallucinations and other symptoms of psychosis in LBDs.³⁰ The 5-HT_{2A}R antagonist clozapine and inverse agonist pimavanserin have both proven effective at treating hallucinations in patients with Parkinson's disease psychosis (PDP).^{31–33} Pimavanserin was recently approved for treatment of PDP by the United States Food and Drug Administration.²⁰⁸

Of particular relevance to this review, SP administration has proved useful for exploring the effect of pimavanserin-modulated 5-HT_{2A}R activation in animal models of Parkinson's disease. McFarland et al.²⁰⁹ employed the head twitch response (HTR), a motor action and key assay of 5-HT_{2A}R activation,^{210,211} which may

index hallucinations in mice. Parkinsonian mice exhibited more spontaneous HTRs, possibly signifying increased LBD-induced hallucinations. They also showed more HTR following SP-administration. Pimavanserin reduced both spontaneous and SP-induced HTRs.

5-HT_{2A}R Upregulation in LBDs

Why is the serotonin system implicated in LBD hallucinations? There is evidence that the 5-HT_{2A}R is upregulated in hallucinating patients.^{212,213} In a postmortem study of PD patients, Huot et al.²¹⁴ reported nearly 50% more 5-HT_{2A}R binding in the inferolateral temporal cortex in patients who experienced VH compared to those that did not. In a PET study, Ballanger et al.²¹⁵ observed increased 5-HT_{2A}R binding in the ventral visual pathway and in frontal cortices among hallucinating PD patients.

5-HT_{2A}R upregulation is also observed in rodent models of PD.^{216–220} There is evidence this effect is a homeostatic response to the loss of dopaminergic function in these animals. Li et al.²²¹ found that reduced dopaminergic function in a PD animal model led to greater serotonergic activity and 5-HT_{2A}R upregulation. Stimulating 5-HT_{2A}R with the psychedelic DOI rescued motor dysfunction, while blocking it worsened symptoms. These findings suggest 5-HT_{2A}R upregulation may compensate for dopaminergic loss

in PD, aiding motor function. Thus, LBD hallucinations may result from complex interactions between dopaminergic and serotonergic systems (see [Box 3](#)).

Excitation in Frontal Cortex It has been proposed that hyperexcitability in the medial prefrontal cortex (mPFC) contributes to LBD hallucinations.³⁰ PD animal models with surgically-induced dopaminergic dysfunction show increased mPFC excitability.^{222,223} Moreover, in the same animals, the excitatory effect of SP administration in mPFC is suppressed. This further suggests that complex dopaminergic and serotonergic interactions are involved in LBD hallucinogenesis (see [Box 3](#)).

The effects of SPs are also linked to hyperexcitability in mPFC.^{131,224,225} In animal models, both DOI^{226,227} and 5-MeO-DMT²²⁸ cause an overall excitatory effect in mPFC. This effect results from changes in E/I balance in frontal cortex, which is comodulated by the 5-HT_{2A}R and 5-HT_{1A}R receptors.^{229,230} Such shifts in E/I balance have long been thought to contribute to hallucinogenesis in both psychosis and psychedelics.^{34,231–233}

Suppression in Visual Cortex Frontal excitability is only half the story, however. Emerging evidence suggests that serotonin primarily suppresses activity in the visual cortex. In a recent optogenetic experiment,

Box 3. Complex Neuromodulator Interactions Contribute to Hallucinogenesis

This review focuses on the link between the serotonergic system and VH, due to its clear role in SP hallucinations.²⁹ However, serotonergic, dopaminergic, and cholinergic systems are closely intertwined in LBDs. Dopaminergic neurodegeneration is a hallmark of LBDs,^{190–192} while degeneration in cholinergic output regions, such as the pedunculopontine nucleus¹⁹³ and basal forebrain,¹⁹⁴ is linked to LBD hallucinations. Dopaminergic medications can worsen LBD hallucinations,^{195–197} whereas cholinergic medications can improve them.¹⁹⁸ Finally, Perry et al.¹⁹⁹ found that hallucinating DLB patients showed severe cholinergic dysfunction but stable dopaminergic and serotonergic function, while non-hallucinators exhibited the opposite pattern.

According to predictive coding frameworks, cholinergic and dopaminergic systems may differentially modulate bottom-up and top-down factors. Cholinergic function enhances sensory precision by up-weighting prediction errors in response to unexpected sensory data, prompting further exploration and updating of internal models (ie, perception).^{200–202} Cholinergic *dysfunction* then, degrades sensory precision and disrupts perceptual updating, ultimately discounting sensory data (ie, bottom-up factors). This failure to reconcile sensory evidence with faulty internal models (ie, hallucinations) may contribute to hallucinatory *persistence*.¹¹³ In contrast, dopaminergic processes modulate high-level priors through prediction errors.^{203,204} Hyperactivation of this system boosts confidence in high-level priors, shifting influence over perceptual updating away from sensory data toward amplified expectations (ie, top-down factors). Fluctuating confidence in these expectations may influence hallucinatory *transitions*.

Teasing apart bottom-up and top-down influences of interacting neuromodulatory systems is a complex goal. Making progress requires aligning task-based and comparative neuropharmacology paradigms across clinical and preclinical experiments.¹³⁵ Some efforts along these lines have been made. For instance, Schmack et al.²⁰⁴ used human psychophysics, pharmacological manipulation, and rodent optogenetics to show that dopaminergic neurons in mice causally increase high-confidence false alarms (ie, “hallucination-like” percepts) in an auditory signal detection task. Anticholinergic drugs that reliably induce hallucinations (eg, scopolamine)²⁰⁵ also increase false alarms.²⁰⁶ It remains unclear whether SPs likewise modulate false-alarm rates.²⁰⁷ Cross-species studies that combine dopaminergic, cholinergic, and serotonergic manipulations in signal detection paradigms or other neurocognitive tasks would significantly clarify how these systems contribute to hallucinatory mechanisms.

Azimi et al.¹⁸⁰ demonstrated that selectively driving either the 5-HT_{1A}R or 5-HT_{2A}R suppresses activity in mouse V1. Driving 5-HT_{1A}R selectively suppresses spontaneous (but not visually evoked) activity, while driving the 5-HT_{2A}R selectively suppresses visually evoked (but not spontaneous) activity. This impressive double dissociation is at least partially consistent with the effects of SPs in mice. Administering DOI, a 5-HT_{2A}R (but not 5-HT_{1A}R) agonist, suppresses visually evoked (but not spontaneous) activity in mouse V1 (see **Figure 2D**).¹⁴⁴

Seillier et al.¹⁸¹ showed that serotonin suppresses visually evoked activity in primate V1 (**Figure 3A**) in a manner similar to 5-HT_{2A}R-dependent visual suppression in mouse V1 (**Figure 2D**). Although the observation made by Seillier et al. does not directly implicate the 5-HT_{2A}R in primate visual suppression, it is worth noting that this receptor is especially densely expressed in primate V1.

More narrowly, 5-HT_{2A}R expression is highly localized to *layer IVc* of the primary visual cortex: the input layer carrying retinal information from thalamus to cortex (**Figure 3B**). This location at the base of the visual cortical hierarchy is uniquely suited for modulating processes throughout the visual system.²³⁶

Two questions present themselves: (A) Does 5-HT_{2A}R activity suppress thalamic input to layer IVc of V1 in primates? (B) Does this putative suppression causally contribute to SP hallucinations? In human research, SP-induced visual suppression and hallucinations are clearly correlated: blocking the 5-HT_{2A}R blocks both effects (**Figure 2B**).¹⁴³ However, this could mean they share a common cause (eg, hyperexcitability), rather than visual suppression directly contributing to hallucinations. Addressing points A and B would determine whether visual degradation plays a mechanistic role in SP hallucinogenesis.¹¹³

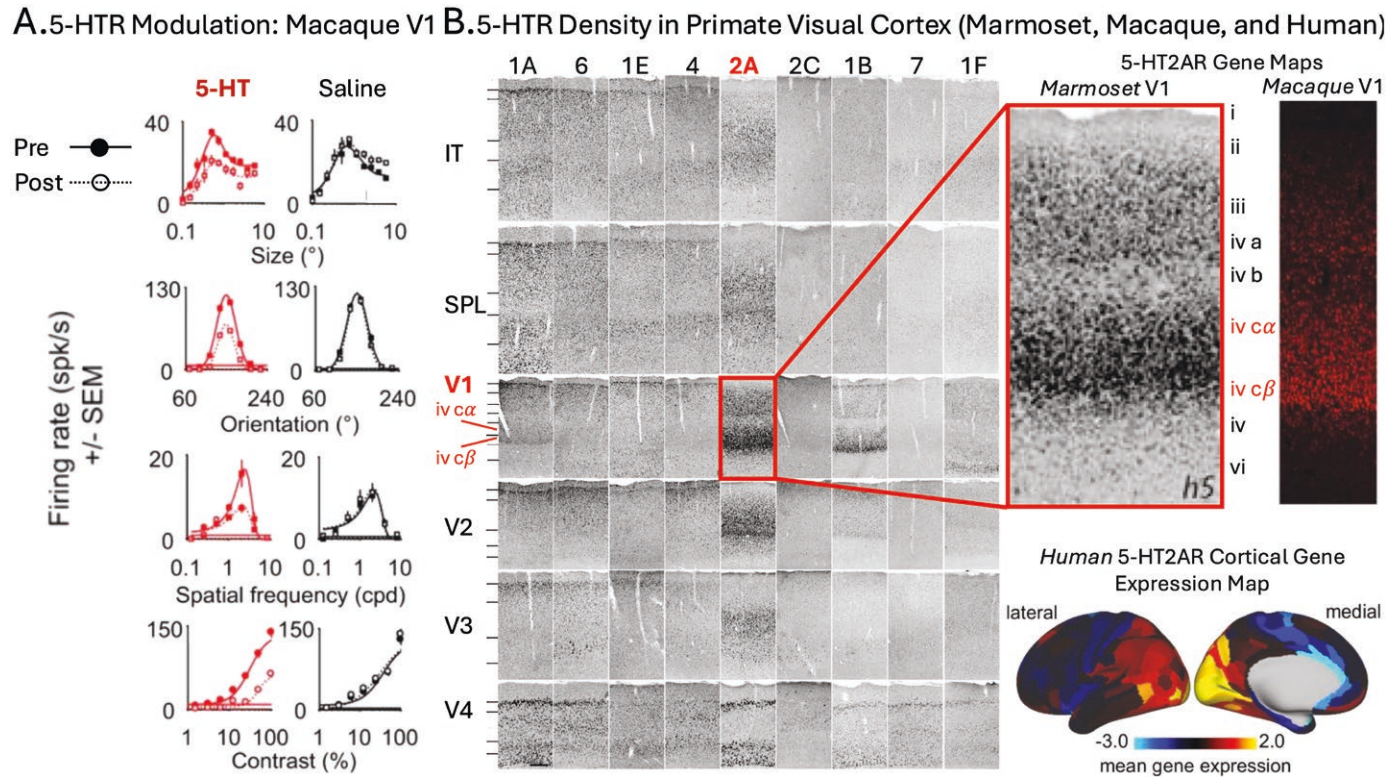


Figure 3. Serotonin Receptor (5-HTR) Modulation and Location in Primate Visual Cortex. (A) From Seillier et al.,¹⁸¹ showing effects of 5-HT iontophoresis in macaque primary visual cortex (V1) during visual stimulation. Compared with saline (NaCl, right panel column), 5-HT administration (dashed-open lines, left panel column) systematically decreased response amplitude of tuning curves for size, orientation, spatial frequency, and contrast stimuli. (B) From Shukla et al.,²³⁴ Watakabe et al.,²³⁵ and Preller et al.,⁹⁸ showing 5-HTR Density in visual cortex of marmoset, macaque, and human, respectively. *Left panel:* Distribution of serotonin receptors derived from in situ hybridization (ISH) across marmoset cortical areas (V1; extrastriate cortex: V2-4; superior parietal lobule: SPL; inferior temporal: IT), showing dense gene expression of 5-HT_{2A}R receptors in geniculorecipient layers IVc α/β of V1. *Upper right panels:* A magnified view of the marmoset V1 5-HT_{2A}R receptor distribution (left) compared side-by-side with macaque V1 receptor images (ISH), illustrating a similar concentration of 5-HT_{2A}R receptors localized to V1 input layers IVc α/β . *Lower right panel:* Human 5-HT_{2A}R gene expression map derived from microarray transcriptional profiling (courtesy of the Allen Human Brain Atlas), showing the cortical distribution of 5-HT_{2A}R mRNA expression. The map highlights the dense 5-HT_{2A}R expression in human primary visual cortex. *Source:* Data from Seillier et al., 2017, and Shukla et al., 2015, are reproduced under a CC BY 4.0 license. Data from Watakabe et al., 2009, is reproduced under a CC BY-NC 2.0 UK license. Data from Preller et al., 2018, is reproduced under a CC BY 4.0 license from *eLife*.

Box 4. Future Directions**Phenomenology**

- [1] Psychophysical experiments on visual distortions common to LBDs and SPs could reveal shared neural mechanisms.
- [2] Interviewing LBD patients who experienced SP-induced hallucinations prior to their diagnosis could clarify phenomenological similarities and the role of expectation in shaping content.
- [3] Quantifying hallucination subtypes in SPs could clarify their prevalence and improve cross-group comparisons.

Cortical Function

- [1] Using consistent neurocognitive tasks (eg, signal detection, ambiguous stimuli, visual working memory) across hallucinatory contexts could pinpoint top-down processes driving specific hallucinations.
- [2] Probing suppressed visual evoked activity and E/I balance shifts could illuminate the role of homeostatic plasticity in hallucinogenesis.
- [3] SP capture experiments could isolate hallucinatory onset and resolution signatures.

Serotonin Receptors

- [1] Combining neuropharmacology with neurocognitive tasks in cross-species studies could clarify serotonergic, dopaminergic, and cholinergic roles in hallucinations.
- [2] Primate studies could test if 5-HT_{2A}R activity suppresses thalamic input to V1.
- [3] Parkinsonian animal models could reveal whether serotonergic visual suppression contributes to LBD cortical neurodegeneration.

While 5-HT_{2A}R-driven visual suppression may contribute to SP-induced hallucinations directly, its role in LBD hallucinations is less clear. One possibility is that *chronic* serotonergic suppression influences cortical degeneration in visual regions, leading to LBD hallucinogenesis. Vignando et al.¹⁰⁷ found that 5-HT_{2A}R /5-HT_{1A}R receptor distributions correlate with hallucination-specific cortical degeneration patterns, distinct from those linked to dopamine. In a potentially related finding, Zarkali et al.²³⁷ reported that the distribution of 5-HT_{2A}R is linked to altered dynamic transitions between integrated and segregated functional connectivity states that are characteristic of hallucinating PD patients. The specific interactions between 5-HT_{2A}R topography, functional connectivity states, and patterns of cortical degeneration in hallucinating patients remain unclear. Future studies using both neuroimaging with patients and Parkinsonian model animals should explore how serotonin's suppressive effects in visual regions might contribute to these structural and functional patterns across the cortical visual hierarchy.

Conclusion

In this review, we compared visual hallucinatory phenomena in LBDs and SPs, proposing that focused research on shared hallucinatory features may help identify

common neural mechanisms. We discussed three cortical signatures found in both LBDs and SPs: hypo-/hyperactivation in sensory/associative regions, reduced visual evoked activity, and a shift in E/I balance toward excitation. We highlight the role of 5-HT_{2A}R /5-HT_{1A}R in hallucinations, suggesting that 5-HT_{2A}R --driven suppression at the base of the cortical visual hierarchy is a potential hallucinogenic mechanism. In SPs, transient visual suppression may trigger compensatory excitability in the associative visual cortex or higher-order regions, contributing to hallucinogenesis. In LBDs, chronic suppression may cause visual-hierarchy degeneration, fostering persistent hallucination proneness. Leveraging SP-induced hallucinations, task-based paradigms, and comparative neuropharmacology across populations and species could reveal shared mechanisms, driving theoretical and clinical advances. Insights into these mechanisms may clarify the fundamental sensory effects of SP administration²³⁶ and guide novel interventions for LBD hallucination, potentially improving clinical management and patient quality of life.

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Conflicts of Interest

FSB is a scientific advisor for MindState Design Labs, LLC and WavePaths, Ltd, and has provided consultation for Gilgamesh Pharmaceuticals, Inc.; KHP is currently an employee of Boehringer Ingelheim GmbH & Co KG.

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